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Star Maker

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PREFACE

AT a moment when Europe is in danger of a catastrophe worse than that of 1914 a book like this may be condemned as a distraction from the desperately urgent defence of civilization against modern barbarism.

Year by year, month by month, the plight of our fragmentary and precarious civilization becomes more serious. Fascism abroad grows more bold and ruthless in its foreign ventures, more tyrannical toward its own citizens, more barbarian in its contempt for the life of the mind. Even in our own country we have reason to

fear a tendency toward militarization and the curtailment of civil liberty. Moreover, while the decades pass, no resolute step is taken to alleviate the injustice of our social order. Our outworn economic system dooms millions to frustration.

In these conditions it is difficult for writers to pursue their calling at once with courage and with balanced judgment. Some merely shrug their shoulders and withdraw from the central struggle of our age. These, with their minds closed against the world's most vital issues, inevitably produce works which not only have no depth of significance for their contemporaries but also are subtly insincere. For these writers must consciously or unconsciously contrive to persuade themselves either that the crisis in human affairs does not exist, or that it is less important than their own work, or that it is anyhow not their business. But the crisis does exist, is of supreme importance, and concerns us all. Can anyone who is at all intelligent and informed hold the contrary without self-deception?

Yet I have a lively sympathy with some of those "intellectuals" who declare that they have no useful contribution to make to the struggle, and therefore had better not dabble in it. I am, in fact, one of them. In our defense I should say that, though we are inactive or ineffective as direct supporters of the cause, we do not ignore it. Indeed, it constantly, obsessively, holds our attention. But we are convinced by prolonged trial and error that the most useful service open to us is indirect. For some writers the case is different. Gallantly plunging into the struggle, they use their powers to spread urgent propaganda, or they even take up arms in the cause. If they have suitable ability, and if the particular struggle in which they serve is in fact a part of the great enterprise of defending (or creating) civilization, they may, of course, do valuable work. In addition they may gain great wealth of experience and human sympathy, thereby immensely increasing their literary power. But the very urgency of their service may tend to blind them to the importance of maintaining and extending, even in this age of crisis, what may be called metaphorically the "self-critical self-consciousness of the human species," or the attempt to see man's life as a whole in relation to the rest of things. This involves the will to regard all human affairs and ideals and theories with as little human prejudice as possible. Those who are in the thick of the struggle inevitably tend to become, though in a great and just cause, partisan. They nobly forgo something of that detachment, that power of cold assessment, which is, after all, among the most valuable human capacities. In their case this is perhaps as it should be; for a desperate struggle demands less of detachment than of devotion. But some who have the cause at heart must serve by striving to maintain, along with human loyalty, a more dispassionate spirit. And perhaps the attempt to see our turbulent world against a background of stars may, after all, increase, not lessen the significance of the present human crisis. It may also strengthen our charity toward one another.

In this belief I have tried to construct an imaginative sketch of the dread but vital whole of things. I know well that it is a ludicrously inadequate and in some ways a childish sketch, even when regarded from the angle of contemporary human experience. In a calmer and a wiser age it might well seem crazy. Yet in spite of its crudity, and in spite of its remoteness, it is perhaps not wholly irrelevant.

At the risk of raising thunder both on the Left and on the Right, I have occasionally used certain ideas and words derived from religion, and I have tried to interpret them in relation to modern needs. The valuable, though much damaged words "spiritual" and "worship," which have become almost as obscene to the Left as the good old sexual words are to the Right, are here intended to suggest an experience which the Right is apt to pervert and the Left to misconceive. This experience, I should say, involves detachment from all private, all social, all racial ends; not in the sense that it leads a man to reject them, but that it makes him prize them in a new way. The "spiritual life" seems to be in essence the attempt to discover and adopt the attitude which is in fact appropriate to our experience as a whole, just as admiration is felt to be in fact appropriate toward a well-grown human being. This enterprise can lead to an increased lucidity and finer temper of consciousness, and therefore can have a great and beneficial effect on behavior. Indeed, if this supremely humanizing experience does not produce, along with a kind of piety toward fate, the resolute will to serve our waking humanity, it is a mere sham and a snare.

Before closing this preface I must express my gratitude to Professor L. C. Martin, Mr. L. H. Myers, and Mr. E. V. Rieu, for much helpful and sympathetic criticism, in consequence of which I rewrote many chapters. Even now I hesitate to associate their names with such an extravagant work. Judged by the standards of the Novel, it is remarkably bad. In fact, it is no novel at all.

Certain ideas about artificial planets were suggested by Mr. J. D. Bernal's fascinating little book The World, the Flesh, and the Devil. I hope he will not strongly disapprove of my treatment of them.

My wife I must thank both for work on the proofs and for being herself.

At the end of the book I have included a note on Magnitude, which may be helpful to readers unfamiliar with astronomy. The very sketchy time scales may amuse some.

O. S. March 1937

CHAPTER I

THE EARTH

1. THE STARTING POINT

ONE night when I had tasted bitterness I went out on to the hill. Dark heather checked my feet. Below marched the suburban lamps. Windows, their curtains drawn, were shut eyes, inwardly watching the lives of dreams. Beyond the sea's level darkness a lighthouse pulsed. Overhead, obscurity. I distinguished our own house, our islet in the tumultuous and bitter currents of the world. There, for a decade and a half, we two, so different in quality, had grown in and in to one another, for mutual support and nourishment, in intricate symbiosis. There daily we planned our several undertakings, and recounted the day's oddities and vexations. There letters piled up to be answered, socks to be darned. There the children were born, those sudden new lives. There, under that roof, our own two lives, recalcitrant sometimes to one another, were all the while thankfully one, one larger, more conscious life than either alone.

All this, surely, was good. Yet there was bitterness. And bitterness not only invaded us from the world; it welled up also within our own magic circle. For horror at our futility, at our own unreality, and not only at the world's delirium, had driven me out on to the hill.

We were always hurrying from one little urgent task to another, but the upshot was insubstantial. Had we, perhaps, misconceived our whole existence? Were we, as it were, living from false premises? And in particular, this partnership of ours, this seemingly so well-based fulcrum for activity in the world, was it after all nothing but a little eddy of complacent and ingrown domesticity, ineffectively whirling on the surface of the great flux, having in itself no depth of being, and no significance? Had we perhaps after all deceived ourselves? Behind those rapt windows did we, like so many others, in-deed live only a dream? In a sick world even the hale are sick. And we two, spinning our little life mostly by rote, sel-dom with clear cognizance, seldom with firm intent, were products of a sick world.

Yet this life of ours was not all sheer and barren fantasy. Was it not spun from the actual fibres of reality, which we gathered in with all the comings and goings through our door, all our traffic with the suburb and the city and with remoter cities, and with the ends of the earth? And were we not spinning together an authentic expression of our own nature? Did not our life issue daily as more or less firm threads of active living, and mesh itself into the growing web, the intricate, ever-proliferating pattern of mankind?

I considered "us" with quiet interest and a kind of amused awe. How could I describe our relationship even to myself without either disparaging it or insulting it with the tawdry decoration of sentimentality? For this our delicate balance of dependence and independence, this coolly critical, shrewdly ridiculing, but loving mutual contact, was surely a microcosm of true community, was after all in its simple style an actual and living example of that high goal which the world seeks.

The whole world? The whole universe? Overhead, obscurity unveiled a star. One tremulous arrow of light, projected how many thousands of years ago, now stung my nerves with vision, and my heart with fear. For in such a universe as this what significance could there be in our fortuitous, our frail, our evanescent community?

But now irrationally I was seized with a strange worship, not, surely of the star, that mere furnace which mere distance falsely sanctified, but of something other, which the dire contrast of the star and us signified to the heart. Yet what, what could thus be signified? Intellect, peering beyond the star, discovered no Star Maker, but only darkness; no Love, no Power even, but only Nothing. And yet the heart praised.

Impatiently I shook off this folly, and reverted from the inscrutable to the familiar and the concrete. Thrusting aside worship, and fear also and bitterness, I determined to examine more coldly this remarkable "us," this surprisingly impressive datum, which to ourselves remained basic to the universe, though in relation to the stars it appeared so slight a thing.

Considered even without reference to our belittling cosmical background, we were after all insignificant, perhaps ridiculous. We were such a commonplace occurrence, so trite, so respectable. We were just a married couple, making shift to live together without undue strain. Marriage in our time was suspect. And ours, with its trivial romantic origin, was doubly suspect.

We had first met when she was a child. Our eyes encountered. She looked at me for a moment with quiet attention; even, I had romantically imagined, with obscure, deep-lying recognition. I, at any rate, recognized in that look (so I persuaded myself in my fever of adolescence) my destiny. Yes! How predestinate had seemed our union! Yet now, in retrospect, how accidental! True, of course, that as a long-married couple we fitted rather neatly, like two close trees whose trunks have grown upwards together as a single shaft, mutually distorting, but mutually supporting. Coldly I now assessed her as merely a useful, but often infuriating adjunct to my personal life. We were on the whole sensible companions. We left one another a certain freedom, and so we were able to endure our proximity.

Such was our relationship. Stated thus it did not seem very significant for the understanding of the universe. Yet in my heart I knew that it was so. Even the cold stars, even the whole cosmos with all its inane immensities could not convince me that this our prized atom of community, imperfect as it was, short-lived as it must be, was not significant.

But could this indescribable union of ours really have any significance at all beyond itself? Did it, for instance, prove that the essential nature of all human beings was to love, rather than to hate and fear? Was it evidence that all men and women the world over, though circumstance might prevent them, were at heart capable of supporting a world-wide, love-knit community? And further, did it, being itself a product of the cosmos, prove that love was in some way basic to the cosmos itself? And did it afford, through its own felt intrinsic excellence, some guarantee that we two, its frail supporters, must in some sense have eternal life? Did it, in fact, prove that love was God, and God awaiting us in his heaven?

No! Our homely, friendly, exasperating, laughter-making, undecorated though most prized community of spirit proved none of these things. It was no certain guarantee of anything but its own imperfect rightness. It was nothing but a very minute, very bright epitome of one out of the many potentialities of existence. I remembered the swarms of the unseeing stars. I remembered the tumult of hate and fear and bitterness which is man's world. I remembered, too, our own not infrequent discordancy. And I reminded myself that we should very soon vanish like the flurry that a breeze has made on still water.

Once more there came to me a perception of the strange contrast of the stars and us. The incalculable potency of the cosmos mysteriously enhanced the Tightness of our brief spark of community, and of mankind's brief, uncertain venture. And these in turn quickened the cosmos.

I sat down on the heather. Overhead obscurity was now in full retreat. In its rear the freed population of the sky sprang out of hiding, star by star.

On every side the shadowy hills or the guessed, featureless sea extended beyond sight. But the hawk-flight of imagination followed them as they curved downward below the horizon. I perceived that I was on a little round grain of rock and metal, filmed with water and with air, whirling in sunlight and darkness. And on the skin of that little grain all the swarms of men, generation by generation, had lived in labor and blindness, with intermittent joy and intermittent lucidity of spirit. And all their history, with its folk-wanderings, its empires, its philosophies, its proud sciences, its social revolutions, its increasing hunger for community, was but a

flicker in one day of the lives of stars.

If one could know whether among that glittering host there were here and there other spirit-inhabited grains of rock and metal, whether man's blundering search for wisdom and for love was a sole and insignificant tremor, or part of a universal movement!

1. EARTH AMONG THE STARS

Overhead obscurity was gone. From horizon to horizon the sky was an unbroken spread of stars. Two planets stared, unwinking. The more obtrusive of the constellations asserted their individuality. Orion's four-square shoulders and feet, his belt and sword, the Plough, the zigzag of Cassiopeia, the intimate Pleiades, all were duly patterned on the dark. The Milky Way, a vague hoop of light, spanned the sky.

Imagination completed what mere sight could not achieve. Looking down, I seemed to see through a transparent planet, through heather and solid rock, through the buried graveyards of vanished species, down through the molten flow of basalt, and on into the Earth's core of iron; then on again, still seemingly downwards, through the southern strata to the southern ocean and lands, past the roots of gum trees and the feet of the inverted antipodeans, through their blue, sun-pierced awning of day, and out into the eternal night, where sun and stars are together. For there, dizzyingly far below me, like fishes in the depth of a lake, lay the nether constellations. The two domes of the sky were fused into one hollow sphere, star-peopled, black, even beside the blinding sun. The young moon was a curve of incandescent wire. The completed hoop of the Milky Way encircled the universe. In a strange vertigo, I looked for reassurance at the little glowing windows of our home. There they still were; and the whole suburb, and the hills. But stars shone through all. It was as though all terrestrial things were made of glass, or of some more limpid, more ethereal vitreosity. Faintly the church clock chimed for midnight. Dimly, receding, it tolled the first stroke.

Imagination was now stimulated to a new, strange mode of perception. Looking from star to star, I saw the heaven no longer as a jeweled ceiling and floor, but as depth beyond flashing depth of suns. And though for the most part the great and familiar lights of the sky stood forth as our near neighbors, some brilliant stars were seen to be in fact remote and mighty, while some dim lamps were visible only because they were so near. On every side the middle distance was crowded with swarms and streams of stars. But even these now seemed near; for the Milky Way had receded into an incomparably greater distance. And through gaps in its nearer parts appeared vista beyond vista of luminous mists, and deep perspectives of stellar populations.

The universe in which fate had set me was no spangled chamber, but a perceived vortex of star-streams. No! It was more. Peering between the stars into the outer darkness, I saw also, as mere flecks and points of light, other such vortices, such galaxies, sparsely scattered in the void, depth beyond depth, so far afield that even the eye of imagination could find no limits to the cosmical, the all-embracing galaxy of galaxies. The universe now appeared to me as a void wherein floated rare flakes of snow, each flake a universe.

Gazing at the faintest and remotest of all the swarm of universes, I seemed, by hypertelescopic imagination, to see it as a population of suns; and near one of those suns was a planet, and on that planet's dark side a hill, and on that hill myself. For our astronomers assure us that in this boundless finitude which we call the cosmos the straight lines of light lead not to infinity but to their source. Then I remembered that, had my vision depended on physical light, and not on the light of imagination, the rays coming thus to me "round" the cosmos would have revealed, not myself, but events that had ceased long before the Earth, or perhaps even the Sun, was formed.

But now, once more shunning these immensities, I looked again for the curtained windows of our home, which, though star-pierced, was still more real to me than all the galaxies. But our home had vanished, with the whole suburb, and the hills too, and the sea. The very ground on which I had been sitting was gone. Instead there lay far below me an insubstantial gloom. And I myself was seemingly disembodied, for I could

neither see nor touch my own flesh. And when I willed to move my limbs, nothing happened. I had no limbs. The familiar inner perceptions of my body, and the headache which had oppressed me since morning, had given way to a vague lightness and exhilaration.

When I realized fully the change that had come over me, I wondered if I had died, and was entering some wholly unexpected new existence. Such a banal possibility at first exasperated me. Then with sudden dismay I understood that if indeed I had died I should not return to my prized, concrete atom of community. The violence of my distress shocked me. But soon I comforted myself with the thought that after all I was probably not dead, but in some sort of trance, from which I might wake at any minute. I resolved, therefore, not to be unduly alarmed by this mysterious change. With scientific interest I would observe all that happened to me.

I noticed that the obscurity which had taken the place of the ground was shrinking and condensing. The nether stars were no longer visible through it. Soon the earth below me was like a huge circular table-top, a broad disc of darkness surrounded by stars. I was apparently soaring away from my native planet at incredible speed. The sun, formerly visible to imagination in the nether heaven, was once more physically eclipsed by the Earth. Though by now I must have been hundreds of miles above the ground, I was not troubled by the absence of oxygen and atmospheric pressure. I experienced only an increasing exhilaration and a delightful effervescence of thought. The extraordinary brilliance of the stars excited me. For, whether through the absence of obscuring air, or through my own increased sensitivity, or both, the sky had taken on an unfamiliar aspect. Every star had seemingly flared up into higher magnitude. The heavens blazed. The major stars were like the headlights of a distant car. The Milky Way, no longer watered down with darkness, was an encircling, granular river of light.

Presently, along the Planet's eastern limb, now far below me, there appeared a faint line of luminosity; which, as I continued to soar, warmed here and there to orange and red. Evidently I was traveling not only upwards but eastwards, and swinging round into the day. Soon the sun leapt into view, devouring the huge crescent of dawn with its brilliance. But as I sped on, sun and planet were seen to drift apart, while the thread of dawn thickened into a misty breadth of sunlight. This increased, like a visibly waxing moon, till half the planet was illuminated. Between the areas of night and day, a belt of shade, warm-tinted, broad as a sub-continent, now marked the area of dawn. As I continued to rise and travel eastwards, I saw the lands swing westward along with the day, till I was over the Pacific and high noon. The Earth appeared now as a great bright orb hundreds of times larger than the full moon. In its center a dazzling patch of light was the sun's image reflected in the ocean. The planet's circumference was an indefinite breadth of luminous haze, fading into the surrounding blackness of space. Much of the northern hemisphere, tilted somewhat toward me, was an expanse of snow and cloud-tops. I could trace parts of the outlines of Japan and China, their vague browns and greens indenting the vague blues and grays of the ocean. Toward the equator, where the air was clearer, the ocean was dark. A little whirl of brilliant cloud was perhaps the upper surface of a hurricane. The Philippines and New Guinea were precisely mapped. Australia faded into the hazy southern limb.

The spectacle before me was strangely moving. Personal anxiety was blotted out by wonder and admiration; for the sheer beauty of our planet surprised me. It was a huge pearl, set in spangled ebony. It was nacrous, it was an opal. No, it was far more lovely than any jewel. Its patterned coloring was more subtle, more ethereal. It displayed the delicacy and brilliance, the intricacy and harmony of a live thing. Strange that in my remoteness I seemed to feel, as never before, the vital presence of Earth as of a creature alive but tranced and obscurely yearning to wake.

I reflected that not one of the visible features of this celestial and living gem revealed the presence of man. Dis-played before me, though invisible, were some of the most congested centers of human population. There below me lay huge industrial regions, blackening the air with smoke. Yet all this thronging life and humanly momentous enterprise had made no mark whatever on the features of the planet. From this high look-out the Earth would have appeared no different before the dawn of man. No visiting angel, or explorer from another

planet, could have guessed that this bland orb teemed with vermin, with world-mastering, self-torturing, incipiently angelic beasts.

CHAPTER 2

INTERSTELLAR TRAVEL

WHILE I was thus contemplating my native planet, I continued to soar through space. The Earth was visibly shrinking into the distance, and as I raced eastwards, it seemed to be rotating beneath me. All its features swung westwards, till presently sunset and the Mid-Atlantic appeared upon its eastern limb, and then the night. Within a few minutes, as it seemed to me, the planet had become an immense half-moon. Soon it was a misty, dwindling crescent, beside the sharp and minute crescent of its satellite.

With amazement I realized that I must be traveling at a fantastic, a quite impossible rate. So rapid was my progress that I seemed to be passing through a constant hail of meteors. They were invisible till they were almost abreast of me; for they shone only by reflected sunlight, appearing for an instant only, as streaks of light, like lamps seen from an express train. Many of them I met in head-on collision, but they made no impression on me. One huge irregular bulk of rock, the size of a house, thoroughly terrified me. The illuminated mass swelled before my gaze, displayed for a fraction of a second a rough and lumpy surface, and then engulfed me. Or rather, I infer that it must have engulfed me; but so swift was my passage that I had no sooner seen it in the middle distance than I found myself already leaving it behind.

Very soon the Earth was a mere star. I say soon, but my sense of the passage of time was now very confused. Minutes and hours, and perhaps even days, even weeks, were now indistinguishable.

While I was still trying to collect myself, I found that I was already beyond the orbit of Mars, and rushing across the thoroughfare of the asteroids. Some of these tiny planets were now so near that they appeared as great stars streaming across the constellations. One or two revealed gibbous, then crescent forms before they faded behind me.

Already Jupiter, far ahead of me, grew increasingly bright and shifted its position among the fixed stars. The great globe now appeared as a disc, which soon was larger than the shrinking sun. Its four major satellites were little pearls floating beside it. The planet's surface now appeared like streaky bacon, by reason of its cloud-zones. Clouds fogged its whole circumference. Now I drew abreast of it and passed it. Owing to the immense depth of its atmosphere, night and day merged into one another without assignable boundary. I noted here and there on its eastern and unilluminated hemisphere vague areas of ruddy light, which were perhaps the glow cast upwards through dense clouds by volcanic upheavals.

In a few minutes, or perhaps years, Jupiter had become once more a star, and then was lost in the splendor of the diminished but still blazing sun. No other of the outer planets lay near my course, but I soon realized that I must be far beyond the limits of even Pluto's orbit. The sun was now merely the brightest of the stars, fading behind me.

At last I had time for distress. Nothing now was visible but the starry sky. The Plough, Cassiopeia, Orion, the Pleiades, mocked me with their familiarity and their remoteness. The sun was now but one among the other bright stars. Nothing changed. Was I doomed to hang thus for ever out in space, a bodiless view-point? Had I died? Was this my punishment for a singularly ineffectual life? Was this the penalty of an inveterate will to remain detached from human affairs and passions and prejudices?

In imagination I struggled back to my suburban hilltop. I saw our home. The door opened. A figure came out into the garden, lit by the hall light. She stood for a moment looking up and down the road, then went back into the house. But all this was imagination only. In actuality, there was nothing but the stars.

After a while I noticed that the sun and all the stars in his neighborhood were ruddy. Those at the opposite pole of the heaven were of an icy blue. The explanation of this strange phenomenon flashed upon me. I was

still traveling, and traveling so fast that light itself was not wholly indifferent to my passage. The overtaking undulations took long to catch me. They therefore affected me as slower pulsations than they normally were, and I saw them therefore as red. Those that met me on my headlong flight were congested and shortened, and were seen as blue.

Very soon the heavens presented an extraordinary appearance, for all the stars directly behind me were now deep red, while those directly ahead were violet. Rubies lay behind me, amethysts ahead of me. Surrounding the ruby constellations there spread an area of topaz stars, and round the amethyst constellations an area of sapphires. Beside my course, on every side, the colors faded into the normal white of the sky's familiar diamonds. Since I was traveling almost in the plane of the galaxy, the hoop of the Milky Way, white on either hand, was violet ahead of me, red behind. Presently the stars immediately before and behind grew dim, then vanished, leaving two starless holes in the heaven, each hole surrounded by a zone of colored stars. Evidently I was still gathering speed. Light from the forward and the hinder stars now reached me in forms beyond the range of my human vision.

As my speed increased, the two starless patches, before and behind, each with its colored fringe, continued to encroach upon the intervening zone of normal stars which lay abreast of me on every side. Amongst these I now detected movement. Through the effect of my own passage the nearer stars appeared to drift across the background of the stars at greater distance. This drifting accelerated, till, for an instant, the whole visible sky was streaked with flying stars. Then everything vanished. Presumably my speed was so great in relation to the stars that light from none of them could take normal effect on me.

Though I was now perhaps traveling faster than light itself, I seemed to be floating at the bottom of a deep and stagnant well. The featureless darkness, the complete lack of all sensation, terrified me, if I may call "terror" the repugnance and foreboding which I now experienced without any of the bodily accompaniments of terror, without any sensation of trembling, sweating, gasping or palpitation. Forlornly, and with self-pity, I longed for home, longed to see once more the face that I knew best. With the mind's eye I could see her now, sitting by the fire sewing, a little furrow of anxiety between her brows. Was my body, I wondered, lying dead on the heather? Would they find it there in the morning? How would she confront this great change in her life? Certainly with a brave face; but she would suffer.

But even while I was desperately rebelling against the dissolution of our treasured atom of community, I was aware that something within me, the essential spirit within me, willed very emphatically not to retreat but to press on with this amazing voyage. Not that my longing for the familiar human world could for'.a moment be counterbalanced by the mere craving for adventure. I was of too home-keeping a kind to seek serious danger and discomfort for their own sake. But timidity was overcome by a sense of the opportunity that fate was giving me, not only to explore the depths of the physical universe, but to discover what part life and mind were actually playing among the stars. A keen hunger now took possession of me, a hunger not for adventure but for insight into the significance of man, or of any manlike beings in the cosmos. This homely treasure of ours, this frank and spring-making daisy beside the arid track of modern life, impelled me to accept gladly my strange adventure; for might I not discover that the whole universe was no mere place of dust and ashes with here and there a stunted life, but actually beyond the parched terrestrial waste land, a world of flowers?

Was man indeed, as he sometimes desired to be, the growing point of the cosmical spirit, in its temporal aspect at least? Or was he one of many million growing points? Or was mankind of no more importance in the universal view than rats in a cathedral? And again, was man's true function power, or wisdom, or love, or worship, or all of all these? Or was the idea of function, of purpose, meaningless in relation to the cosmos? These grave questions I would answer. Also I must learn to see a little more clearly and confront a little more rightly (so I put it to myself) that which, when we glimpse it at all, compels our worship.

I now seemed to my self-important self to be no isolated individual, craving aggrandizement, but rather an emissary of mankind, no, an organ of exploration, a feeler, 'projected by the living human world to make

contact with its fellows in space. At all cost I must go forward, even if my trivial earthly life must come to an untimely end, and my wife and children be left without me. I must go forward; and somehow, some day, even if after centuries of interstellar travel, I must return.

When I look back on that phase of exaltation, now that I have indeed returned to earth after the most bewildering adventures, I am dismayed at the contrast between the spiritual treasure which I aspired to hand over to my fellow men and the paucity of my actual tribute. This failure was perhaps due to the fact that, though I did indeed accept the challenge of the adventure, I accepted it only with secret reservations. Fear and the longing for comfort, I now recognize, dimmed the brightness of my will. My resolution, so boldly formed, proved after all frail. My unsteady courage often gave place to yearnings for my native planet. Over and over again in the course of my travels I had a sense that, owing to my timid and pedestrian nature, I missed the most significant aspects of events.

Of all that I experienced on my travels, only a fraction was clearly intelligible to me even at the time; and then, as I shall tell, my native powers were aided by beings of superhuman development. Now that I am once more on my native planet, and this aid is no longer available, I cannot recapture even so much of the deeper insight as I formerly attained. And so my record, which tells of the most far-reaching of all human explorations, turns out to be after all no more reliable than the rigmarole of any mind unhinged by the impact of experience beyond its comprehension.

To return to my story. How long I spent in debate with myself I do not know, but soon after I had made my decision, the absolute darkness was pierced once more by the stars. I was apparently at rest, for stars were visible in every direction, and their color was normal.

But a mysterious change had come over me. I soon discovered that, by merely willing to approach a star, I could set myself in motion toward it, and at such a speed that I must have traveled much faster than normal light. This, as I knew very well, was physically impossible. Scientists had assured me that motion faster than the speed of light was meaningless. I inferred that my motion must therefore be in some manner a mental, not a physical phenomenon, that I was enabled to take up successive viewpoints without physical means of locomotion. It seemed to me evident, too, that the light with which the stars were now revealed to me was not normal, physical light; for I noticed that my new and expeditious means of travel took no effect upon the visible colors of the stars. However fast I moved, they retained their diamond hues, though all were somewhat brighter and more tinted than in normal vision.

No sooner had I made sure of my new power of locomotion than I began feverishly to use it. I told myself that I was embarking on a voyage of astronomical and metaphysical research; but already my craving for the Earth was distorting my purpose. It turned my attention unduly toward the search for planets, and especially for planets of the terrestrial type.

At random I directed my course toward one of the brighter of the near stars. So rapid was my advance that certain lesser and still nearer luminaries streamed past me like meteors. I swung close to the great sun, insensitive to its heat. On its mottled surface, in spite of the pervading brilliance, I could see, with my miraculous vision, a group of huge dark sun-spots, each one a pit into which a dozen Earths could have been dropped. Round the star's limb the excrescences of the chromosphere looked like fiery trees and plumes and prehistoric monsters, atiptoe or awing, all on a globe too small for them. Beyond these the pale corona spread its films into the darkness. As I rounded the star in hyperbolic flight I searched anxiously for planets, but found none. I searched again, meticulously, tacking and veering near and far. In the wider orbits a small object like the earth might easily be overlooked. I found nothing but meteors and a few insubstantial comets. This was the more disappointing because the star seemed to be of much the same type as the familiar sun. Secretly I had hoped to discover not merely planets but actually the Earth.

Once more I struck out into the ocean of space, heading for another near star. Once more I was disappointed. I

approached yet another lonely furnace. This too was unattended by the minute grains that harbor life.

I now hurried from star to star, a lost dog looking for its master. I rushed hither and thither, intent on finding a sun with planets, and among those planets my home. Star after star I searched, but far more I passed impatiently, recognizing at once that they were too large and tenuous and young to be Earth's luminary. Some were vague ruddy giants broader than the orbit of Jupiter; some, smaller and more definite, had the brilliance of a thousand suns, and their color was blue. I had been told that our Sun was of average type, but I now discovered many more of the great youngsters than of the shrunken, yellowish middle-aged. Seemingly I must have strayed into a region of late stellar condensation.

I noticed, but only to avoid them, great clouds of dust, huge as constellations, eclipsing the star-streams; and tracts of palely glowing gas, shining sometimes by their own light, sometimes by the reflected light of stars. Often these nacrous cloud-continents had secreted within them a number of vague pearls of light, the embryos of future stars. I glanced heedlessly at many star-couples, trios, and quartets, in which more or less equal partners waltz in close union. Once, and once only, I came on one of those rare couples in which one partner is no bigger than a mere Earth, but massive as a whole great star, and very brilliant. Up and down this region of the galaxy I found here and there a dying star, somberly smoldering; and here and there the encrusted and extinguished dead. These I could not see till I was almost upon them, and then only dimly, by the reflected light of the whole heaven. I never approached nearer to them than I could help, for they were of no interest to me in my crazy yearning for the Earth. Moreover, they struck a chill into my mind, prophesying the universal death. I was comforted, however, to find that as yet there were so few of them.

I found no planets. I knew well that the birth of planets was due to the close approach of two or more stars, and that such accidents must be very uncommon. I reminded myself that stars with planets must be as rare in the galaxy as gems among the grains of sand on the sea-shore. What chance had I of coming upon one? I began to lose heart. The appalling desert of darkness and barren fire, the huge emptiness so sparsely pricked with scintillations, the colossal futility of the whole universe, hideously oppressed me. And now, an added distress, my power of locomotion began to fail. Only with a great effort could I move at all among the stars, and then but slowly, and ever more slowly. Soon I should find myself pinned fast in space like a fly in a collection; but lonely, eternally alone. Yes, surely this was my special Hell.

I pulled myself together. I reminded myself that even if this was to be my fate, it was no great matter. The Earth could very well do without me. And even if there was no other living world anywhere in the cosmos, still, the Earth itself had life, and might wake to far fuller life. And even though I had lost my native planet, still, that beloved world was real. Besides, my whole adventure was a miracle, and by continued miracle might I not stumble on some other Earth? I remembered that I had undertaken a high pilgrimage, and that I was man's emissary to the stars.

With returning courage my power of locomotion returned. Evidently it depended on a vigorous and self-detached mentality. My recent mood of self-pity and earthward-yearning had hampered it.

Resolving to explore a new region of the galaxy, where perhaps there would be more of the older stars and a greater hope of planets, I headed in the direction of a remote and populous cluster. From the faintness of the individual members of this vaguely speckled ball of light I guessed that it must be very far afield. On and on I traveled in the darkness. As I never turned aside to search, my course through the ocean of space never took me near enough to any star to reveal it as a disc. The lights of heaven streamed remotely past me like the lights of distant ships. After a voyage during which I lost all measure of time I found myself in a great desert, empty of stars, a gap between two star-streams, a cleft in the galaxy. The Milky Way surrounded me, and in all directions lay the normal dust of distant stars; but there were no considerable lights, save the thistle-down of the remote cluster which was my goal.

This unfamiliar sky disturbed me with a sense of my increasing dissociation from my home. It was almost a

comfort to note, beyond the furthest stars of our galaxy, the minute smudges that were alien galaxies, incomparably more distant than the deepest recesses of the Milky Way; and to be reminded that, in spite of all my headlong and miraculous traveling, I was still within my native galaxy, within the same little cell of the cosmos where she, my life's friend, still lived. I was surprised, by the way, that so many of the alien galaxies appeared to the naked eye, and that the largest was a pale, cloudy mark bigger than the moon in the terrestrial sky.

By contrast with the remote galaxies, on whose appearance all my voyaging failed to make impression, the star-cluster ahead of me was now visibly expanding. Soon after I had crossed the great emptiness between the star-streams, my cluster confronted me as a huge cloud of brilliants. Presently I was passing through a more populous area, and then the cluster itself opened out ahead of me, covering the whole forward sky with its congested lights. As a ship approaching port encounters other craft, so I came upon and passed star after star. When I had penetrated into the heart of the cluster, I was in a region far more populous than any that I had explored. On every side the sky blazed with suns, many of which appeared far brighter than Venus in the Earth's sky. I felt the exhilaration of a traveler who, after an ocean crossing, enters harbors by night and finds himself surrounded by the lights of a metropolis. In this congested region, I told myself, many close approaches must have occurred, many planetary systems must have been formed. Once more I looked for middle-aged stars of the sun's type. All that I had passed hitherto were young giants, great as the whole solar system. After further searching I found a few likely stars, but none had planets. I found also many double and triple stars, describing their incalculable orbits; and great continents of gas, in which new stars were condensing. At last, at last I found a planetary system. With almost insupportable hope I circled among these worlds; but all were greater than Jupiter, and all were molten. Again I hurried from star to star. I must have visited thousands, but all in vain. Sick and lonely I fled out of the cluster. It dwindled behind me into a ball of down, sparkling with dew-drops. In front of me a great tract of darkness blotted out a section of the Milky Way and the neighboring area of stars, save for a few near lights which lay between me and the obscuring opacity. The billowy edges of this huge cloud of gas or dust were revealed by the glancing rays of bright stars beyond it. The sight moved me with self-pity; on so many nights at home had I seen the edges of dark clouds silvered just so by moonlight. But the cloud which now opposed me could have swallowed not merely whole worlds, not merely countless planetary systems, but whole constellations.

Once more my courage failed me. Miserably I tried to shut out the immensities by closing my eyes. But I had neither eyes nor eyelids. I was a disembodied, wandering view-point. I tried to conjure up the little interior of my home, with the curtains drawn and the fire dancing. I tried to persuade myself that all this horror of darkness and dis tance and barren incandescence was a dream, that I was dozing by the fire, that at any moment I might wake, that she would reach over from her sewing and touch me and smile. But the stars still held me prisoner.

Again, though with failing strength, I set about my search. And after I had wandered from star to star for a period that might have been days or years or aeons, luck or some guardian spirit directed me to a certain sun-like star; and looking outwards from this center, I caught sight of a little point of light, moving, with my movement, against the patterned sky.4 As I leapt toward it, I saw another, and another. Here was indeed a planetary system much like my own. So obsessed was I with human standards that I sought out at once the most earth-like of these worlds. And amazinsly earth-like it appeared, as its disc swelled before me, or below me. Its atmosphere was evidently less dense than ours, for the outlines of unfamiliar continents and oceans were very plainly visible.

As on the earth, the dark sea brilliantly reflected the sun's image. White cloud-tracts lay here and there over the seas and the lands, which, as on my own planet, were mottled green and brown. But even from this height I saw that the greens were more vivid and far more blue than terrestrial vegetation. I noted, also, that on this planet there was less ocean than land, and that the centers of the great continents were chiefly occupied by dazzling creamy-white deserts.

CHAPTER 3

THE OTHER EARTH

1. ON THE OTHER EARTH

AS I slowly descended toward the surface of the little planet, I found myself searching for a land which promised to be like England. But no sooner did I realize what I was doing than I reminded myself that conditions here would be entirely different from terrestrial conditions, and that it was very unlikely that I should find intelligent beings at all. If such beings existed, they would probably be quite incomprehensible to me. Perhaps they would be huge spiders or creeping jellies. How could I hope ever to make contact with such monsters?

After circling about at random for some time over the filmy clouds and the forests, over the dappled plains and prairies and the dazzling stretches of desert, I selected a maritime country in the temperate zone, a brilliantly green peninsula. When I had descended almost to the ground, I was amazed at the verdure of the country-side. Here unmistakably was vegetation, similar to ours in essential character, but quite unfamiliar in detail. The fat, or even bulbous, leaves reminded me of our desert-flora, but here the stems were lean and wiry. Perhaps the most striking character of this vegetation was its color, which was a vivid blue-green, like the color of vineyards that have been treated with copper salts. I was to discover later that the plants of this world had indeed learnt to protect themselves by means of copper sulphate from the microbes and the insect-like pests which formerly devastated this rather dry planet.

I skimmed over a brilliant prairie scattered with Prussian blue bushes. The sky also attained a depth of blue quite unknown on earth, save at great altitudes. There were a few low yet cirrus clouds, whose feathery character I took to be due to the tenuousness of the atmosphere. This was borne out by the fact that, though my descent had taken place in the forenoon of a summer's day, several stars managed to pierce the almost nocturnal sky. All exposed surfaces were very intensely illuminated. The shadows of the nearer bushes were nearly black. Some distant objects, rather like buildings, but probably mere rocks, appeared to be blocked out in ebony and snow. Altogether the landscape was one of unearthly and fantastical beauty.

I glided with wingless flight over the surface of the planet, through glades, across tracts of fractured rock, along the banks of streams. Presently I came to a wide region covered by neat, parallel rows of fern-like plants, bearing masses of nuts on the lower surfaces of their leaves. It was almost impossible to believe that this vegetable regimentation had not been intelligently planned. Or could it after all be merely a natural phenomenon not known on my own planet? Such was my surprise that my power of locomotion, always subject to emotional interference, now began to fail me. I reeled in the air like a drunk man. Pulling myself together, I staggered on over the ranked crops toward a rather large object which lay some distance from me beside a strip of bare ground. Presently, to my amazement, my stupefaction, this object revealed itself as a plow. It was rather a queer instrument, but there was no mistaking the shape of the blade, which was rusty, and obviously made of iron. There were two iron handles, and chains for attachment to a beast of burden. It was difficult to believe that I was many light-years distant from England. Looking round, I saw an unmistakable cart track, and a bit of dirty ragged cloth hanging on a bush. Yet overhead was the unearthly sky, full noon with stars.

I followed the lane through a little wood of queer bushes, whose large fat drooping leaves had cherry-like fruits along their edges. Suddenly, round a bend in the lane, I came upon a man. Or so at first he seemed to my astounded and star-weary sight. I should not have been so surprised by the strangely human character of this creature had I at this early stage understood the forces that controlled my adventure. Influences which I shall later describe doomed me to discover first such worlds as were most akin to my own. Meanwhile the reader may well conceive my amazement at this strange encounter. I had always supposed that man was a unique being. An inconceivably complex conjunction of circumstances had produced him, and it was not to be

supposed that such conditions would be repeated anywhere in the universe. Yet here, on the very first globe to be explored, was an obvious peasant. Approaching him, I saw that he was not quite so like terrestrial man as he seemed at a distance; but he was a man for all that. Had God, then, peopled the whole universe with our kind? Did he perhaps in very truth make us in his image? It was incredible. To ask such questions proved that I had lost my mental balance.

As I was a mere disembodied view-point, I was able to observe without being observed. I floated about him as he strode along the lane. He was an erect biped and in general plan definitely human. I had no means of judging his height, but he must have been approximately of normal terrestrial stature, or at least not smaller than a pigmy and not taller than a giant. He was of slender build. His legs were almost like a bird's, and enclosed in rough narrow trousers. Above the waist he was naked, displaying a disproportionately large thorax, shaggy with greenish hair. He had two short but powerful arms, and huge shoulder muscles. His skin was dark and ruddy, and dusted plentifully with bright green down. All his contours were uncouth, for the details of muscles, sinews and joints were very plainly different from our own. His neck was curiously long and supple. His head I can best describe by saying that most of the brain-pan, covered with a green thatch, seemed to have slipped backwards and downwards over the nape. His two very human eyes peered from under the eaves of hair. An oddly projecting, almost spout-like mouth made him look as though he were whistling. Between the eyes, and rather above them, was a pair of great equine nostrils which were constantly in motion. The bridge of the nose was represented by an elevation in the thatch, reaching from the nostrils backwards over the top of the head. There were no visible ears. I discovered later that the auditory organs opened into the nostrils.

Clearly, although evolution on this Earth-like planet must have taken a course on the whole surprisingly like that which had produced my own kind, there must also have been many divergencies.

The stranger wore not only boots but gloves, seemingly ol tough leather. His boots were extremely short. I was to discover later that the feet of this race, the "Other Men," as I called them, were rather like the feet of an ostrich or a camel. The instep consisted of three great toes grown together. In place of the heel there was an additional broad, stumpy toe. The hands were without palms. Each was a bunch of three gristly fingers and a thumb.

The aim of this book is not to tell of ray own adventures but to give some idea of the worlds which I visited. I shall therefore not recount in detail how I established myself among the Other Men. Of myself it is enough to say a few words. When I had studied this agriculturalist for a while, I began to be strangely oppressed by his complete unawareness, of myself. With painful clearness I realized that the purpose of my pilgrimage was not merely scientific observation, but also the need to effect some kind of mental and spiritual traffic with other worlds, for mutual enrichment and community. How should I ever be able to achieve this end unless I could find some means of communication? It was not until I had followed my companion to his home, and had spent many days in that little circular stone house with roof of mudded wicker, that I discovered the power of entering into his mind, of seeing through his eyes, sensing through all his sense organs, perceiving his world just as he perceived it, and following much of his thought and his emotional life. Not till very much later, when I had passively "inhabited" many individuals of the race, did I discover how to make my presence known, and even to converse inwardly with my host.

This kind of internal "telepathic" intercourse, which was to serve me in all my wanderings, was at first difficult, ineffective, and painful. But in time I came to be able to live through the experiences of my host with vividness and accuracy, while yet preserving my own individuality, my own critical intelligence, my own desires and fears. Only when the other had come to realize my presence within him could he, by a special act of volition, keep particular thoughts secret from me.

It can well be understood that at first I found these alien minds quite unintelligible. Their very sensations differed from my familiar sensations in important respects. Their thoughts and all their emotions and

sentiments were strange to me. The traditional groundwork of these minds, their most familiar concepts, were derived from a strange history, and expressed in languages which to the terrestrial mind were subtly misleading.

I spent on the Other Earth many "other years," wandering from mind to mind and country to country, but I did not gain any clear understanding of the psychology of the Other Men and the significance of their history till I had encountered one of their philosophers, an aging but still vigorous man whose eccentric and unpalatable views had prevented him from attaining eminence. Most of my hosts, when they became aware of 'my presence within them, regarded me either as an evil spirit or as a divine messenger. The more sophisticated, however, assumed that I was a mere disease, a symptom of insanity in themselves. They therefore promptly applied to the local "Mental Sanitation Officer." After I had spent, according to the local calendar, a year or so of bitter loneliness among minds who refused to treat me as a human being, I had the good fortune to come under the philosopher's notice. One of my hosts, who complained of suffering from "voices," and visions of "another world," appealed to the old man for help. Bvalltu, for such approximately was the philosopher's name, the "11" being pronounced more or less as in Welsh, Bvalltu effected a "cure" by merely inviting me to accept the hospitality of his own mind, where, he said, he would very gladly entertain me. It was with extravagant joy that I made contact at last with a being who recognized in me a human personality.

2. A BUSY WORLD

So many important characteristics of this world-society need to be described that I cannot spend much time on the more obvious features of the planet and its race. Civilization had reached a stage of growth much like that which was familiar to me. I was constantly surprised by the blend of similarity and difference. Traveling over the planet I found that cultivation had spread over most of the suitable areas, and that industrialism was already far advanced in many countries. On the prairies huge flocks of mammal-like creatures grazed and scampered. Larger mammals, or quasi-mammals, were farmed on all the best pasture land for food and leather. I say "quasi-mammal" because, though these creatures were viviparous, they did not suckle. The chewed cud, chemically treated in the maternal belly, was spat into the offspring's mouth as a jet of pre-digested fluid. It was thus also that human mothers fed their young.

The most important means of locomotion on the Other Earth was the steam-train, but trains in this world were so bulky that they looked like whole terraces of houses on the move. This remarkable railway development was probably due to the great number and length of journeys across deserts. Occasionally I traveled on steam-ships on the few and small oceans, but marine transport was on the whole backward. The screw propeller was unknown, its place being taken by paddle wheels. Internal-combustion engines were used in road and desert transport. Flying, owing to the rarified atmosphere, had not been achieved; but rocket-propulsion was already used for long-distance transport of mails, and for long-range bombardment in war. Its application to aeronautics might come any day.

My first visit to the metropolis of one of the great empires of the Other Earth was an outstanding experience. Everything was at once so strange and so familiar. There were streets and many-windowed stores and offices. In this old city the streets were narrow, and so congested was the motor traffic that pedestrians were accommodated on special elevated tracks slung beside the first-story windows and across the streets.

The crowds that streamed along these footpaths were as variegated as our own. The men wore cloth tunics, and trousers surprisingly like the trousers of Europe, save that the crease affected by the respectable was at the side of the leg. The women, breastless and high-nostriled like the men, were to be distinguished by their more tubular lips, whose biological function it was to project food for the infant. In place of skirts they disported green and glossy silk tights and little gawdy knickers. To my unaccustomed vision the effect was inexpressibly vulgar. In summar both sexes often appeared in the streets naked to the waist; but they always wore gloves.

Here, then, was a host of persons who, in spite of their oddity, were as essentially human as Londoners. They went about their private affairs with complete assurance, ignorant that a spectator from another world found them one and all grotesque, with their lack of forehead, their great elevated quivering nostrils, their startlingly human eyes, their spout-like mouths. There they were, alive and busy, shopping, staring, talking. Children dragged at their mothers' hands. Old men with white facial hair bowed over walking-stocks. Young men eyed young women. The prosperous were easily to be distinguished from the unfortunate by their newer and richer clothes, their confident and sometimes arrogant carriage.

How can I describe in a few pages the distinctive character of a whole teeming and storied world, so different from my own, yet so similar? Here, as on my own planet, infants were being born every hour. Here, as there, they clamored for food, and very soon for companionship. They discovered what pain was, and what fear, and what loneliness, and love. They grew up, molded by the harsh or kindly pressure of their fellows, to be either well nurtured, generous, sound, or mentally crippled, bitter, unwittingly vindictive. One and all they desperately craved the bliss of true community; and very few, fewer here, perhaps, than in my own world, found more than the vanishing flavor of it. They howled with the pack and hounded with the pack. Starved both physically and mentally, they brawled over the quarry and tore one another to pieces, mad with hunger, physical or mental. Sometimes some of them paused and asked what it was all for; and there followed a battle of words, but no clear answer. Suddenly they were old and finished. Then, the span from birth to death being an imperceptible instant of cosmical time, they vanished.

This planet, being essentially of the terrestrial type, had produced a race that was essentially human, though, so to speak, human in a different key from the terrestrial. These continents were as variegated as ours, and inhabited by a race as diversified as Homo sapiens. All the modes and facets of the spirit manifested in our history had their equivalents in .3. the history of the Other Men. As with us, there had been dark ages and ages of brilliance, phases of advancement and of retreat, cultures predominantly material, and others in the main intellectual, aesthetic, or spiritual. There were "Eastern" races and "Western" races. There were empires, republics, dictatorships. Yet all was different from the terrestrial. Many of the differences, of course, were superficial; but there was also an underlying, deep-lying difference which I took long to understand and will not yet describe. I must begin by speaking of the biological equipment of the Other Men. Their animal nature was at bottom much like ours. They responded with anger, fear, hate, tenderness, curiosity, and so on, much as we respond. In sensory equipment they were not unlike ourselves, save that in vision they were less sensitive to color and more to form than is common with us. The violent colors of the Other Earth appeared to me through the eyes of its natives very subdued. In hear-ing also they were rather ill-equipped. Though their auditory organs were as sensitive as ours to faint sounds, they were poor discriminators. Music, such as we know, never developed in this world.

In compensation, scent and taste developed amazingly. These beings tasted not only with their mouths, but with then-moist black hands and with their feet. They were thus afforded an extraordinarily rich and intimate experience of their planet. Tastes of metals and woods, of sour and sweet earths, of the many rocks, and of the innumerable shy or bold flavors of plants crushed beneath the bare running feet, made up a whole world unknown to terrestrial man.

The genitals also were equipped with taste organs. There were several distinctive male and female patterns of chemical characteristics, each powerfully attractive to the opposite sex. These were savored faintly by contact of hands or feet with any part of the body, and with exquisite intensity in copulation.

This surprising richness of gustatory experience made it very difficult for me to enter fully into the thoughts of the Other Men. Taste played as important a part in their imagery and conception as sight in our own. Many ideas which terrestrial man has reached by way of sight, and which even in their most abstract form still bear traces of their visual origin, the Other Men conceived in terms of taste. For example, our "brilliant," as applied to persons or ideas, they would translate by a word whose literal meaning was "tasty." For "lucid" they would use a term which in primitive times was employed by hunters to signify an easily runnable taste-trail. To have

"religious illumination" was to "taste the meadows of heaven." Many of our non-visual concepts also were rendered by means of taste. "Complexity" was "many flavored," a word applied originally to the confusion of tastes round a drinking pool frequented by many kinds of beasts. "Incompatibility" was derived from a word meaning the disgust which certain human types felt for one another on account of their flavors.

Differences of race, which in our world are chiefly conceived in terms of bodily appearance, were for the Other Men almost entirely differences of taste and smell. And as the races of the Other Men were much less sharply localized than our own races, the strife between groups whose flavors were repugnant to one another played a great part in history. Each race tended to believe that its own flavor was characteristic of all the finer mental qualities, was indeed an absolutely reliable label of spiritual worth. In former ages the gustatory and olfactory differences had, no doubt, been true signs of racial differences; but in modern times, and in the more developed lands, there had been great changes. Not only had the races ceased to be clearly localized, but also industrial civilization had produced a crop of genetic changes which rendered the old racial distinctions meaningless. The ancient flavors, however, though they had by now no racial significance at all, and indeed members of one family might have mutually repugnant flavors, continued to have the traditional emotional effects. In each country some particular flavor was considered the true hall-mark of the race of that country, and all other flavors were despised, if not actually condemned.

In the country which I came to know best the orthodox racial flavor was a kind of saltness inconceivable to terrestrial man. My hosts regarded themselves as the very salt of the earth. But as a matter of fact the peasant whom I first "inhabited" was the only genuine pure salt man of orthodox variety whom I ever encountered. The great majority of that country's citizens attained their correct taste and smell by artificial means. Those who were at least approximately salt, with some variety of saltness, though not the ideal variety, were forever exposing the deceit of their sour, sweet, or bitter neighbors. Unfortunately, though the taste of the limbs could be fairly well disguised, no effective means had been found for changing the flavor of copulation. Consequently newly married couples were apt to make the most shattering discoveries about one another on the wedding night. Since in the great majority of unions neither party had the orthodox flavor, both were willing to pretend to the world that all was well. But often there would turn out to be a nauseating incompatibility between the two gustatory types. The whole population was rotten with neuroses bred of these secret tragedies of marriage. Occasionally, when one party was more or less of the orthodox flavor, this genuinely salt partner would indignantly denounce the impostor. The courts, the news bulletins, and the public would then join in self-righteous protests.

Some "racial" flavors were too obtrusive to be disguised. One in particular, a kind of bitter-sweet, exposed its possessor to extravagant persecution in all but the most tolerant countries. In past times the bitter-sweet race had earned a reputation of cunning and self-seeking, and had been periodically massacred by its less intelligent neighbors. But in the general biological ferment of modern times the bitter-sweet flavor might crop up in any family. Woe, then, to the accursed infant, and to all its relatives! Persecution was inevitable; unless indeed the family was wealthy enough to purchase from the state "an honorary salting" (or in the neighboring land, "an honorary sweetening"), which removed the stigma.

In the more enlightened countries the whole racial superstition was becoming suspect. There was a movement among the intelligentsia for conditioning infants to tolerate every kind of human flavor, and for discarding the deodorants and degustatants, and even the boots and gloves, which civilized convention imposed.

Unfortunately this movement of toleration was hampered by one of the consequences of industrialism. In the congested and unhealthy industrial centers a new gustatory and olfactory type had appeared, apparently as a biological mutation. In a couple of generations this sour, astringent, and undisguis-able flavor dominated in all the most disreputable working-class quarters. To the fastidious palates of the well-to-do it was overwhelmingly nauseating and terrifying. In fact it became for them an unconscious symbol, tapping all the secret guilt and fear and hate which the oppressors felt for the oppressed.

In this world, as in our own, nearly all the chief means of production, nearly all the land, mines, factories, railways, ships, were controlled for private profit by a small minority of the population. These privileged individuals were able to force the masses to work for them on pain of starvation. The tragic farce inherent in such a system was already approaching. The owners directed the energy of the workers increasingly toward the production of more means of production rather than to the fulfilment of the needs of individual life. For machinery might bring profit to the owners; bread would not. With the increasing competition of machine with machine, profits declined, and therefore wages, and therefore effective demand for goods. Marketless products were destroyed, though bellies were unfed and backs unclad. Unemployment, disorder, and stem repression increased as the economic system disintegrated. A familiar story!

As conditions deteriorated, and the movements of charity and state-charity became less and less able to cope with the increasing mass of unemployment and destitution, the new pariah-race became more and more psychologically useful to the hate-needs of the scared, but still powerful, prosperous. The theory was spread that these wretched beings were the result of secret systematic race-pollution by riff-raff immigrants, and that they deserved no consideration whatever. They were therefore allowed only the basest forms of employment and the harshest conditions of work. When unemployment had become a serious social problem, practically the whole pariah stock was workless and destitute. It was of course easily believed that unemployment, far from being due to the decline of capitalism, was due to the worthless-ness of the pariahs.

At the time of my visit the working class had become tainted through and through by the pariah stock, and there was a vigorous movement afoot amongst the wealthy and the official classes to institute slavery for pariahs and half-pariahs, so that these might be openly treated as the cattle which in fact they were. In view of the danger of continued race-pollution, some politicians urged wholesale slaughter of the pariahs, or, at the least, universal sterilization. Others pointed out that, as a supply of cheap labor was necessary to society, it would be wiser merely to keep their numbers down by working them to an early death in occupations which those of "pure race" would never accept. This, at any rate, should be done in times of prosperity; but in times of decline, the excess population could be allowed to starve, or might be used up in the physiological laboratories.

The persons who first dared to suggest this policy were scourged by the whips of generous popular indignation. But their policy was in fact adopted; not explicitly but by tacit consent, and in the absence of any more constructive plan.

The first time that I was taken through the poorest quarter of the city I was surprised to see that, though there were large areas of slum property far more squalid than anything in England, there were also many great clean blocks of tenements worthy of Vienna. These were surrounded by gardens, which were crowded with wretched tents and shanties. The grass was worn away, the bushes damaged, the flowers trampled. Everywhere men, women, and children, all filthy and ragged, were idling.

I learned that these noble buildings had been erected before the world-economic-crisis (familiar phrase!) by a millionaire who had made his money in trading an opium-like drug. He presented the buildings to the City Council, and was gathered to heaven by way of the peerage. The more deserving and less unsavory poor were duly housed; but care was taken to fix the rent high enough to exclude the pariah-race. Then came the crisis. One by one the tenants failed to pay their rent, and were ejected. Within a year the buildings were almost empty.

There followed a very curious sequence of events, and one which, as I was to discover, was characteristic of this strange world. Respectable public opinion, though vindictive toward the unemployed, was passionately tender toward the sick. In falling ill, a man acquired a special sanctity, and exercised a claim over all healthy persons. Thus no sooner did any of the wretched campers succumb to a serious disease than he was carried off to be cared for by all the resources of medical science. The desperate paupers soon discovered how things stood, and did all in their power to fall sick. So successful were they, that the hospitals were soon filled. The

empty tenements were therefore hastily fitted out to receive the increasing flood of patients.

Observing these and other farcical events, I was reminded of my own race. But though the Other Men were in many ways so like us, I suspected increasingly that some factor still hidden from me doomed them to a frustration which my own nobler species need never fear. Psychological mechanisms which in our case are tempered with common sense or moral sense stood out in this world in flagrant excess. Yet it was not true that Other Man was less intelligent or less moral than man of my own species. In abstract thought and practical invention he was at least our equal. Many of his most recent advances in physics and astronomy had passed beyond our present attainment. I noticed, however, that psychology was even more chaotic than with us, and that social thought was strangely perverted.

In radio and television, for instance, the Other Men were technically far ahead of us, but the use to which they put their astounding inventions was disastrous. In civilized countries everyone but the pariahs carried a pocket receiving set. As the Other Men had no music, this may seem odd; but since they lacked newspapers, radio was the only means by which the man in the street could learn the lottery and sporting results which were his staple mental diet. The place of music, moreover, was taken by taste- and smell-themes, which were translated into patterns of ethereal undulation, transmitted by all the great national stations, and restored to their original form in the pocket receivers and taste-batteries of the population. These instruments afforded intricate stimuli to the taste organs and scent organs of the hand. Such was the power of this kind of entertainment that both men and women were nearly always seen with one hand in a pocket. A special wave length had been allotted to the soothing of infants.

A sexual receiving set had been put upon the market, and programs were broadcast for it in many countries; but not in all. This extraordinary invention was a combination of radio--touch, taste, odor, and sound. It worked not through the sense organs, but direct stimulation of the appropriate brain-centers. The recipient wore a specially constructed skullcap, which transmitted to him from a remote studio the embraces of some delectable and responsive woman, as they were then actually being experienced by a male "love-broadcaster" or as electromagnetically recorded on a steel tape on some earlier occasion. Controversies had arisen about the morality of sexual broadcasting. Some countries permitted programs for males but not for females, wishing to preserve the innocence of the purer sex. Elsewhere the clerics had succeeded in crushing the whole project on the score that radio-sex, even for men alone, would be a diabolical substitute for a certain much desired and jealously guarded religious experience, called the immaculate union, of which I shall tell in the sequel. Well did the priests know that their power depended largely on their ability to induce this luscious ecstasy in their flock by means of ritual and other psychological techniques.

Militarists also were strongly opposed to the new invention; for in the cheap and efficient production of illusory sexual embraces they saw a danger even more serious than contraception. The supply of cannon-fodder would decline.

Since in all the more respectable countries broadcasting had been put under the control of retired soldiers or good churchmen, the new device was at first adopted only in the more commercial and the more disreputable states. From their broadcasting stations the embraces of popular "radio love-stars" and even of impecunious aristocrats were broadcast along with advertisements of patent medicines, taste-proof gloves, lottery results, savors, and degustatants.

The principle of radio-brain-stimulation was soon developed much further. Programs of all the most luscious or piquant experiences were broadcast in all countries, and could be picked up by simple receivers that were within the means of all save the pariahs. Thus even the laborer and the factory hand could have the pleasures of a banquet without expense and subsequent repletion, the delights of proficient dancing without the trouble of learning the art, the thrills of motor-racing without danger. In an ice-bound northern home he could bask on tropical beaches, and in the tropics indulge in winter sports. Governments soon discovered that the new invention gave them a cheap and effective kind of power over their subjects. Slum-conditions could be

tolerated if there was an unfailing supply of illusory luxury. Reforms distasteful to the authorities could be shelved if they could be represented as inimical to the national radio-system. Strikes and riots could often be broken by the mere threat to close down the broadcasting studios, or alternatively by flooding the ether at a critical moment with some saccharine novelty.

The fact that the political Left Wing opposed the further development of radio amusements made Governments and the propertied classes the more ready to accept it. The Communists, for the dialectic of history on this curiously earth-like planet had produced a party deserving that name, strongly condemned the scheme. In their view it was pure Capitalist dope, calculated to prevent the otherwise inevitable dictatorship of the proletariat.

The increasing opposition of the Communists made it possible to buy off the opposition of their natural enemies, the priests and soldiers. It was arranged that religious services should in future occupy a larger proportion of broadcasting time, and that a tithe of all licensing fees should be allocated to the churches. The offer to broadcast the immaculate union, however, was rejected by the clerics. As an additional concession it was agreed that all married members of the staffs of Broadcasting Authorities must, on pain of dismissal, prove that they had never spent a night away from their wives (or husbands). It was also agreed to weed out all those B.A. employees who were suspected of sympathy with such disreputable ideals as pacifism and freedom of expression. The soldiers were further appeased by a state-subsidy for maternity, a tax on bachelors, and regular broadcasting of military propaganda.

During my last years on the Other Earth a system was invented by which a man could retire to bed for life and spend all his time receiving radio programs. His nourishment and all his bodily functions were attended to by doctors and nurses attached to the Broadcasting Authority. In place of exercise he received periodic massage. Participation in the scheme was at first an expensive luxury, but its inventors hoped to make it at no distant date available to all. It was even expected that in time medical and menial attendants would cease to be necessary. A vast system of automatic food-production, and distribution of liquid pabulum by means of pipes leading to the mouths of the recumbent subjects, would be complemented by an intricate sewage system. Electric massage could be applied at will by pressing a button. Medical supervision would be displaced by an automatic endocrine-compensation system. This would enable the condition of the patient's blood to regulate itself automatically by tapping from the communal drug-pipes whatever chemicals were needed for correct physiological balance.

Even in the case of broadcasting itself the human element would no longer be needed, for all possible experiences would have been already recorded from the most exquisite living examples. These would be continuously broadcast in a great number of alternative programs.

A few technicians and organizers might still be needed to superintend the system; but, properly distributed, their work would entail for each member of the World Broadcasting Authority's staff no more than a few hours of interesting activity each week.

Children, if future generations were required, would be produced ectogenetically. The World Director of Broadcasting would be requested to submit psychological and physiological specifications of the ideal "listening breed." Infants produced in accordance with this pattern would then be educated by special radio programs to prepare them for adult radio life. They would never leave their cots, save to pass by stages to the full-sized beds of maturity. At the latter end of life, if medical science did not succeed in circumventing senility and death, the individual would at least be able to secure a painless end by pressing an appropriate button.

Enthusiasm for this astounding project spread rapidly in all civilized countries, but certain forces of reaction were bitterly opposed to it. The old-fashioned religious people and the militant nationalists both affirmed that it was man's glory to be active. The religious held that only in self-discipline, mortification of the flesh, and